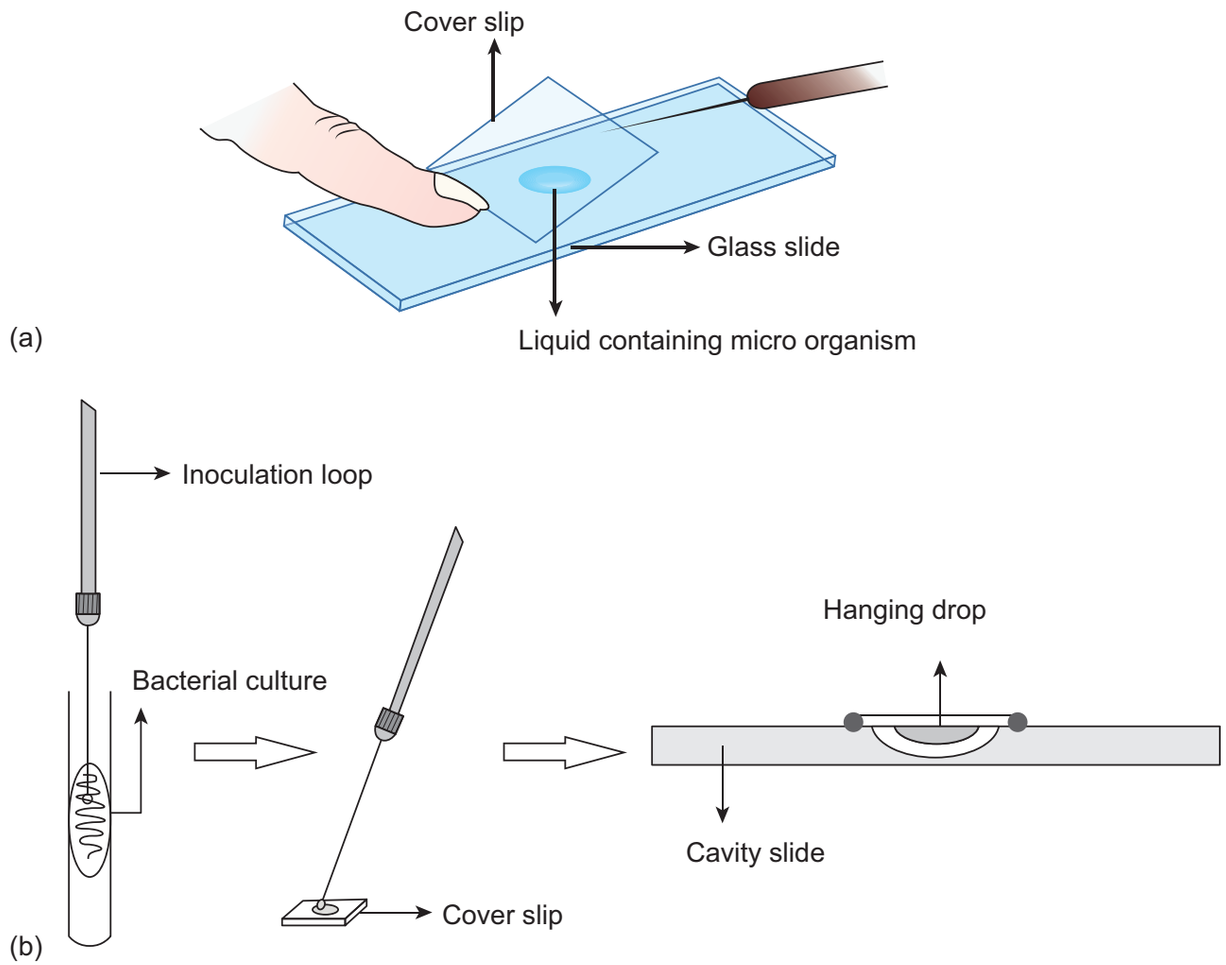



**Figure 3.1:** a) Wet mount and b) Hanging drop preparation

## 3.3 Stains

Stains are dyes used to increase colour contrast. Dye is a coloured organic compound that adheres to microbial cells, giving colour to the cell. Today several stains and staining procedures are available to study the morphological details of various microorganisms. The process of imparting colour to the microbial cell is known as staining.

Stains are organic compounds containing chromophore and auxochrome groups linked to benzene ring.

A chromophore group imparts colour to the compound. Compounds of benzene containing chromophore radicals are called chromogens. Such a compound, even though it is coloured, is not a dye. In order for a compound to be a dye, it must contain not only a chromophore group but also another group known as auxochrome that imparts the property of electrolytic dissociation. Auxochrome gives salt forming properties to the compound.

Hence, each stain or dye is composed of three components:

(i) Benzene ring: It is the basic colourless structural component of a stain or dye.

(ii) Chromophore: It is the functional group that gives colour.

(iii) Auxochrome: It is the group that gives ionic properties to the stain.

The term stain and dye are not the same. The basic differences between dye and stain are given in Table 3.1.



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